High Performance Computing – A practical approach

➢ Pedagogy:

- **Concept → System Understanding → Hands-On:** Each day introduces core HPC concepts followed by practical execution on Linux, oneAPI, GPUs, and cluster environments.
- **Industry-Grade Tools:** Training uses Intel oneAPI, AI Analytics Toolkit, TensorFlow-MKL, Horovod, VTune, Slurm, and OpenVINO—mirroring real HPC workflows.
- **Practical, Cluster-Based Learning:** Learners run ML/DL workloads on HPC nodes, submit jobs with Slurm, and analyze hardware performance.
- **Performance & Optimization Focus:** Emphasis on parallel computing, distributed training, profiling, and hardware-aware optimization.
- **Project-Oriented:** Daily labs build toward end-to-end AI workflows—from ML processing to distributed training and LLM optimization on HPC hardware.

➢ Learning Outcomes:

HPC & Linux Foundations

- Understand HPC architectures, Intel Xeon/GPU/accelerator ecosystems, and secure remote access (SSH).
- Navigate Linux environments, automate tasks with shell scripting, and use oneAPI toolkits.

Optimized ML Workflows

- Build and run parallelized ML models using Intel-optimized NumPy, Pandas, and Scikit-learn.
- Leverage Intel AI Analytics Toolkit for accelerated traditional ML workflows.

Deep Learning on HPC

- Train TensorFlow models optimized with oneDNN/MKL on Xeon CPUs and Intel GPUs.
- Submit, manage, and scale DL jobs using Slurm and distributed frameworks like Horovod.

Distributed AI & Performance Engineering

- Perform distributed TensorFlow training across nodes.
- Use Intel VTune to profile compute-intensive workloads and optimize performance.

Generative AI & LLMs on HPC

- Deploy and fine-tune LLMs using Intel HPC hardware (Gaudi, Xeon).
- Optimize inference with OpenVINO for real-world generative AI applications.

Day 1: Fundamentals of HPC, Linux & Shell Scripting

- Introduction to HPC,
- Intel HPC architecture (Xeon, GPUs, AI accelerators);
- Linux basics: File system, commands, environment setup;
- Shell scripting for automation;
- Secured Shell
- Intel oneAPI toolkit
- Introduction to Intel HPC Toolkit

Day 2: Traditional ML using Intel OneAPI

- Parallel Computing
- OpenMP for multithreading, MPI for distributed systems,
- Data decomposition and synchronization
- Using Intel AI Analytics Toolkit (Optimized Scikit-learn, Pandas, NumPy);
- Intel Extension for Scikit-learn for optimized ML algorithms;

Day 3: Deep Learning on HPC with TensorFlow, Slurm & GPUs

- ML workflows on HPC systems, Optimized libraries
- Introduction to hardware-aware ML optimization
- Intel-optimized TensorFlow (tensorflow-mkl, oneDNN);
- Running DL models on Intel GPUs (Xeon, Habana Gaudi);
- Slurm job scheduling for Deep Learning (batch jobs, resource allocation);
- Training Tensorflow with Horovod
- Distributed training using Horovod

Day 4: Distributed AI Training & Profiling Optimization

- Distributed TensorFlow training with Slurm;
- Profiling & debugging with Intel Vtune

Day 5: Generative AI & LLMs on Intel HPC

- Introduction to Generative AI & LLMs on HPC
- Intel hardware for LLMs (Xeon, Habana Gaudi);
- Optimizing LLM inference with OpenVINO Toolkit;
- Fine-tuning LLMs on HPC